PATTON TANG

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PROFESSIONAL EXPERIENCE

Brooks Automation | Fremont, CA — June 2025 - Present

Software Engineer

- Designed and developed event-driven finite state machine to enable API control of new machine elements under a publish-subscribe architecture in respects to customer specifications in C#. Implemented solution services for over 5,000 customers.
- Implemented bug fixes and software features within an Agile development cycle, collaborating with senior engineers on architecture and tracking via Azure DevOps.
- Built and implemented a CSV file conversion API for clients and service engineers, transforming log data and machine configurations into a human readable format.
- Documented test cases to validate software functionality and identify potential defects, collaborating with SQA and cross-functional engineering teams.

Software Engineer Intern — May 2022 - August 2022

- Expanded preset software configurations to improve customer user experience and shorten SQA testing by multiple hours per session with C#. Tracked with git and TFS.

Freelance Artist | Various, United States — October 2023 – May 2025

- Analyzed and recorded client metrics to develop data-driven recommendations aligned with client goals, evaluating branding, product diversity, inventory, and location.

PROJECTS

TypeFighter | Javascript, React, Flask, Vercel, PostgreSQL

- Co-deployed a typing test website to improve user words per minute via gamification methods such as through traditional score leaderboards and community driven goals.
- Designed system architecture for PostgreSQL database to store Open Library API calls.
- Calculated and displayed real-time words per minute (WPM) and accuracy (%) within 10ms error with safeguards to ensure data authenticity.

TECHNICAL SKILLS

- **Languages**: Python, Java, C++, SQL, C#, Javascript, HTML, CSS
- **Frameworks**: Flask, .NET, React
- **Developer Tools**: Git, Microsoft Suite (Excel, Word, Powerpoint), Visual Studio

EDUCATION

California State University, Fullerton | Fullerton, CA — Aug 2022 - May 2025

Bachelor of Science in Computer Science — GPA: 3.5 / 4.0, cum laude

- Relevant Coursework: Web Front-End & Back-End Engineering, Algorithm Engineering, Data Structures, File Structures and Database Systems, Data Science and Big Data